Applied Machine Learning

19-02-26

Supervised ML

when lebelled is used for model training

eg. weather prediction

$$\underbrace{\begin{matrix}\text{temperature}\\ \text{wind speed}\\ \text{humidity}\end{matrix}}_{\text{features}} \longrightarrow \boxed{\begin{matrix}\text{ML}\\ \text{model}\end{matrix}} \longrightarrow \underbrace{\begin{matrix}\text{possibility of}\\ \text{rain (0-100)}\end{matrix}}_{\text{target}}$$

regression

eg. fruit detection using image

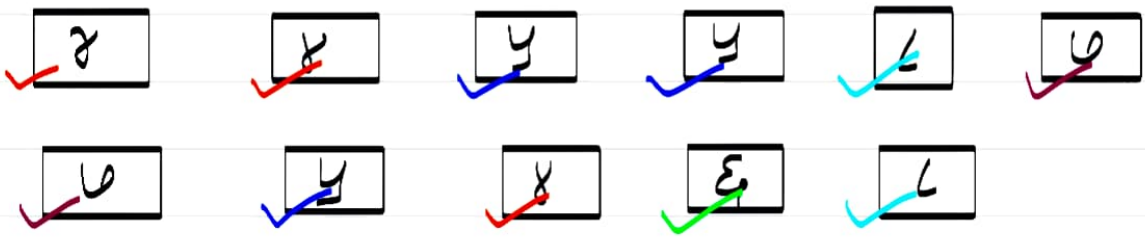
image	label
$\square$	mango
$\square$	banana
$\square$	apple
⋮	⋮

$$\underbrace{\text{image}}_{\text{feature}} \longrightarrow \boxed{\begin{matrix}\text{ML}\\ \text{model}\end{matrix}} \longrightarrow \underbrace{\begin{matrix}\text{apple/ banana/}\\ \text{mango}\end{matrix}}_{\text{target}}$$

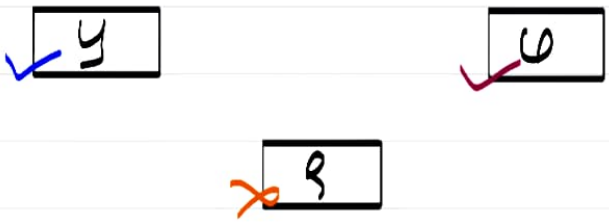
classification

- Sometimes data doesn't have target, it's called unlabelled data.

eg.



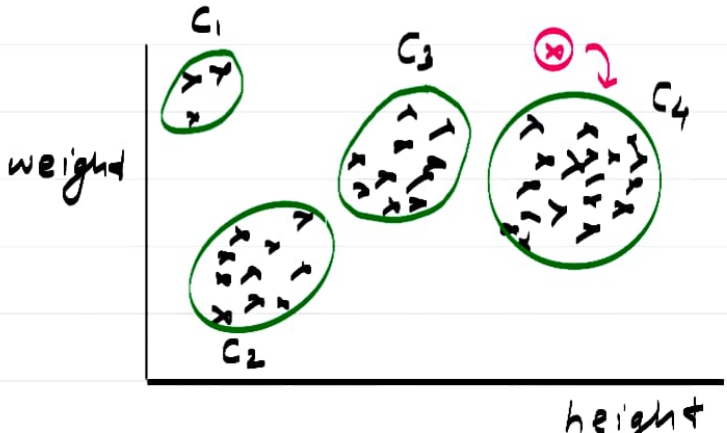
clustering
↓
unsupervised learning
↓
using unlabelled data



Supervised ML	Unsupervised ML
- labelled data	- unlabelled data
- features + target	- only features
- regression, classification	- clustering, PCA

eg.

height	weight
150	50
140	43
175	65
⋮	⋮



- Dimensionality Reduction - PCA

its an unsupervised ML model

eg. housing data

carpet area	parking area (10% of carpet)	rooms	city
100	10	1	Hyd
200	20	2	Hyd
150	15	1	Hyd
300	30	2	Hyd
⋮	⋮	⋮	⋮

this column has single value for all rows hence includes very less information
↓
we can remove it

4 features ⇒ 4 dimensional data

after removing city column

3 features = 3 dimensional data

further, we can create a new column - total area as carpet + parking area

carpet area	parking area	rooms	total area	rooms
100	10	1	110	1
150	15	1	165	1
200	20	2	220	2
300	30	2	330	2
⋮	⋮	⋮	⋮	⋮
∞	∞	3 dimensions		2 dimensions

- dimensionality reduction from 3 to 2

Supervised ML

- labelled data is used (features + target)

features $\longrightarrow$ $\boxed{\text{ML model}}$ $\longrightarrow$ target

eg.

x_1	x_2	y
1	1	3
2	1	4
1	2	5
0	2	4
3	1	5
1	3	7
features		target

$x_1, x_2 \longrightarrow \boxed{\text{model}} \longrightarrow y$

$y = f(x_1, x_2)$

$\boxed{y = x_1 + 2x_2}$ → this function represents relationship between features x_1, x_2 and target y

$\left.\begin{array}{l} x_1 = 1 \\ x_2 = 5 \end{array}\right\} y = 11$

- As the data is very small and relationship is very easy to identify, we could do it manually.

- But when data is huge and relationships are complex, we need supervised ML to automatically identify the relationships for us.

Due to this we can say

- objective of supervised ML is to correctly identify relationship between input and output.

- A ML algorithm / function can easily represent this relationship and the algorithm has to be very accurate.

eg. house price prediction

area	price
100	10
120	12
130	15
⋮	⋮
feature	target

area → $\boxed{\text{model}}$ → price

$y = mx + c$

straight line

price = m × area + c

eg.

area	rooms	city	price

area
rooms
city → $\boxed{\text{model}}$ → price

$$\text{price} = w_1\,\text{area} + w_2\,\text{rooms} + w_3\,\text{city} + c$$

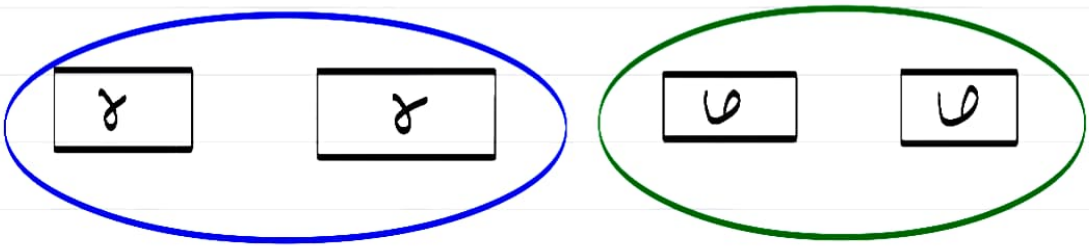
weights = numbers

bias

} Linear regression

Unsupervised ML

- uses unlabelled data
- clustering, dimensionality reduction



- objective of unsupervised ML is to learn the hidden patterns in input features and use those patterns for clustering/dimensionality reduction

ML
- supervised
  - regression
  - classification
- Unsupervised
  - clustering
  - dimensionality reduction

Regression vs classification

features → model → target

- continuous → regression
  - eg. house price prediction, height prediction
- categorical or discrete → classification
  - sentiment analysis, face recognition